

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC: ANTI-LOCK BRAKE SYSTEM CONTROL MODULE [ABS]

[G1794597]

DESCRIPTION AND OPERATION

ABS CONTROL MODULE

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places and with a current calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed via the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMS which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all diagnostic trouble codes (DTCs) that could be logged in the ABS control module, for additional diagnosis and testing information refer to the relevant diagnosis and testing section For additional information, refer to: [Anti-Lock Control - Stability Assist](#) (206-09 Anti-Lock Control - Stability Assist, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0030-38	Left Front Tone Wheel - Signal frequency incorrect	NOTE: Circuit reference - WSP FL / WSS FL - Signal frequency incorrect - Magnetic pulse ring damaged/contaminated - Wheel speed sensor air gap - Wiring harness or wheel speed sensor connector fault - Damaged wheel speed sensor magnetic encoder - Faulty wheel bearing/hub assembly - Incorrect wheel assembly/tire size	- Check for any dust or metal particles stuck to the wheel speed sensor and tone/bearing ring and rectify as required - Check the wheel speed sensor air gap and mating surface for correct sensor seating and rectify as required - Check for evidence of faulty connector/pins backed out. Replace as required - Check for any water ingress to the sensor. Replace as required - Check integrity of the wheel bearing. Replace as required - Check the integrity of the wheel hub assembly. Replace as required - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle. Check the wheel and tire for condition and balance and rectify/replace as required. Check and tighten the wheel nuts - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0031-11	Left Front Wheel Speed Sensor - Circuit short to ground	NOTE: Circuit reference - WSP FL / WSS FL - Wheel speed sensor power supply circuit - Short circuit to ground - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor power supply circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required
C0031-12	Left Front Wheel Speed Sensor - Circuit short to battery	NOTE: Circuit reference - WSP FL - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to power - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to power. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required
C0031-14	Left Front Wheel Speed Sensor - Circuit short to ground or open	NOTE: Circuit reference - WSP FL - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to ground, open circuit, high resistance - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - Further diagnostics may be accessed using the Jaguar Land Rover Approved Diagnostic Equipment to check datalogger signal - Left Front Wheel Speed Sensor Input - (0x2B06)

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0031-1C	Left Front Wheel Speed Sensor - Circuit voltage out of range	NOTE: Circuit reference - WSP FL - Wheel speed sensor circuit - Voltage out of range - Wheel speed sensor circuit - Short circuit to power, short circuit to another sensor, open circuit, high resistance	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to power, short circuit to another sensor, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest
C0031-29	Left Front Wheel Speed Sensor - Signal invalid	NOTE: Circuit reference - WSP FL / WSS FL - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0031-2F	Left Front Wheel Speed Sensor - Signal erratic	NOTE: Circuit reference - WSS FL - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0031-31	Left Front Wheel Speed Sensor - No signal	NOTE: Circuit reference - WSS FL - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0031-4A	Left Front Wheel Speed Sensor - Incorrect component installed	NOTE: Circuit reference - WSP FL / WSS FL Incorrect wheel speed sensor installed	Check and install the correct wheel speed sensor. Clear the DTCs and retest. To extinguish warning lamps cycle the ignition state to off, then return the ignition state to on and wait up to 30 seconds
C0031-62	Left Front Wheel Speed Sensor - Signal compare failure	NOTE: Circuit reference - WSP FL / WSS FL - Wheel speed sensor signal compare failure - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0031-66	Left Front Wheel Speed Sensor - Signal has too many transitions / events	NOTE: Circuit reference - WSP FL / WSS FL - Unexpected wheel speed sensor signal detected - Wheel speed in excess of maximum expected value - Wheel speed sensor circuit - Fault - Incorrect wheel assembly/tire size - Wheel speed sensor fault	- Check the front left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0033-38	Right Front Tone Wheel - Signal frequency incorrect	NOTE: Circuit reference - WSP FR / WSS FR - Signal frequency incorrect - Magnetic pulse ring damaged/contaminated - Wheel speed sensor air gap - Wiring harness or wheel speed sensor connector fault - Damaged wheel speed sensor magnetic encoder - Faulty wheel bearing/hub assembly - Incorrect wheel assembly/tire size	- Check for any dust or metal particles stuck to the wheel speed sensor and tone/bearing ring and rectify as required - Check the wheel speed sensor air gap and mating surface for correct sensor seating and rectify as required - Check for evidence of faulty connector/pins backed out. Replace as required - Check for any water ingress to the sensor. Replace as required - Check integrity of the wheel bearing. Replace as required - Check the integrity of the wheel hub assembly. Replace as required - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle. Check the wheel and tire for condition and balance and rectify/replace as required. Check and tighten the wheel nuts - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0034-11	Right Front Wheel Speed Sensor - Circuit short to ground	NOTE: Circuit reference - WSP FR - Wheel speed sensor power supply circuit - Short circuit to ground - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor power supply circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0034-12	Right Front Wheel Speed Sensor - Circuit short to battery	NOTE: Circuit reference - WSP FR - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to power - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to power. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required
C0034-14	Right Front Wheel Speed Sensor - Circuit short to ground or open	NOTE: Circuit reference - WSP FR - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to ground, open circuit, high resistance - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - Further diagnostics may be accessed using the Jaguar Land Rover Approved Diagnostic Equipment to check datalogger signal - Right Front Wheel Speed Sensor Input - (0x2B07)
C0034-1C	Right Front Wheel Speed Sensor - Circuit voltage out of range	NOTE: Circuit reference - WSP FR - Wheel speed sensor circuit - Voltage out of range - Wheel speed sensor circuit - Short circuit to power, short circuit to another sensor, open circuit, high resistance	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to power, short circuit to another sensor, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest
C0034-29	Right Front Wheel Speed Sensor - Signal invalid	NOTE: Circuit reference - WSS FR - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0034-2F	Right Front Wheel Speed Sensor - Signal erratic	NOTE: Circuit reference - WSS FR - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0034-31	Right Front Wheel Speed Sensor - No signal	NOTE: Circuit reference - WSP FR / WSS FR - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0034-4A	Right Front Wheel Speed Sensor - Incorrect component installed	NOTE: Circuit reference - WSP FR / WSS FR Incorrect wheel speed sensor installed	Check and install the correct wheel speed sensor. Clear the DTCs and retest. To extinguish warning lamps cycle the ignition state to off, then return the ignition state to on and wait up to 30 seconds

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0034-62	Right Front Wheel Speed Sensor - Signal compare failure	NOTE: Circuit reference - WSP FR / WSS FR - Wheel speed sensor signal compare failure - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0034-66	Right Front Wheel Speed Sensor - Signal has too many transitions / events	NOTE: Circuit reference - WSP FR / WSS FR - Unexpected wheel speed sensor signal detected - Wheel speed in excess of maximum expected value - Wheel speed sensor circuit - Fault - Incorrect wheel assembly/tire size - Wheel speed sensor fault	- Check the front right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0036-38	Left Rear Tone Wheel - Signal frequency incorrect	NOTE: Circuit reference - WSP RL / WSS RL - Signal frequency incorrect - Magnetic pulse ring damaged/contaminated - Wheel speed sensor air gap - Wiring harness or wheel speed sensor connector fault - Damaged wheel speed sensor magnetic encoder - Faulty wheel bearing/hub assembly - Incorrect wheel assembly/tire size	- Check for any dust or metal particles stuck to the wheel speed sensor and tone/bearing ring and rectify as required - Check the wheel speed sensor air gap and mating surface for correct sensor seating and rectify as required - Check for evidence of faulty connector/pins backed out. Replace as required - Check for any water ingress to the sensor. Replace as required - Check integrity of the wheel bearing. Replace as required - Check the integrity of the wheel hub assembly. Replace as required - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle. Check the wheel and tire for condition and balance and rectify/replace as required. Check and tighten the wheel nuts - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0037-11	Left Rear Wheel Speed Sensor - Circuit short to ground	NOTE: Circuit reference - WSP RL - Wheel speed sensor power supply circuit - Short circuit to ground - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor power supply circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required
C0037-12	Left Rear Wheel Speed Sensor - Circuit short to battery	NOTE: Circuit reference - WSP RL - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to power - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to power. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required
C0037-14	Left Rear Wheel Speed Sensor - Circuit short to ground or open	NOTE: Circuit reference - WSP RL - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to ground, open circuit, high resistance - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - Further diagnostics may be accessed using the Jaguar Land Rover Approved Diagnostic Equipment to check datalogger signal - Left Rear Wheel Speed Sensor Input - (0x2B08)

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0037-1C	Left Rear Wheel Speed Sensor - Circuit voltage out of range	NOTE: Circuit reference - WSP RL - Wheel speed sensor circuit - Voltage out of range - Wheel speed sensor circuit - Short circuit to power, short circuit to another sensor, open circuit, high resistance	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to power, short circuit to another sensor, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest
C0037-29	Left Rear Wheel Speed Sensor - Signal invalid	NOTE: Circuit reference - WSS RL - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0037-2F	Left Rear Wheel Speed Sensor - Signal erratic	NOTE: Circuit reference - WSS RL - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0037-31	Left Rear Wheel Speed Sensor - No signal	NOTE: Circuit reference - WSS RL - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0037-4A	Left Rear Wheel Speed Sensor - Incorrect component installed	NOTE: Circuit reference - WSP RL / WSS RL Incorrect wheel speed sensor installed	Check and install the correct wheel speed sensor. Clear the DTCs and retest. To extinguish warning lamps cycle the ignition state to off, then return the ignition state to on and wait up to 30 seconds
C0037-62	Left Rear Wheel Speed Sensor - Signal compare failure	NOTE: Circuit reference - WSP RL / WSS RL - Wheel speed sensor signal compare failure - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0037-66	Left Rear Wheel Speed Sensor - Signal has too many transitions / events	NOTE: Circuit reference - WSP RL / WSS RL - Unexpected wheel speed sensor signal detected - Wheel speed in excess of maximum expected value - Wheel speed sensor circuit - Fault - Incorrect wheel assembly/tire size - Wheel speed sensor fault	- Check the rear left wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0039-38	Right Rear Tone Wheel - Signal frequency incorrect	NOTE: Circuit reference - WSP RR / WSS RR - Signal frequency incorrect - Magnetic pulse ring damaged/contaminated - Wheel speed sensor air gap - Wiring harness or wheel speed sensor connector fault - Damaged wheel speed sensor magnetic encoder - Faulty wheel bearing/hub assembly - Incorrect wheel assembly/tire size	- Check for any dust or metal particles stuck to the wheel speed sensor and tone/bearing ring and rectify as required - Check the wheel speed sensor air gap and mating surface for correct sensor seating and rectify as required - Check for evidence of faulty connector/pins backed out. Replace as required - Check for any water ingress to the sensor. Replace as required - Check integrity of the wheel bearing. Replace as required - Check the integrity of the wheel hub assembly. Replace as required - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle. Check the wheel and tire for condition and balance and rectify/replace as required. Check and tighten the wheel nuts - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C003A-11	Right Rear Wheel Speed Sensor - Circuit short to ground	NOTE: Circuit reference - WSP RR - Wheel speed sensor power supply circuit - Short circuit to ground - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor power supply circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C003A-12	Right Rear Wheel Speed Sensor - Circuit short to battery	NOTE: Circuit reference - WSP RR - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to power - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to power. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required
C003A-14	Right Rear Wheel Speed Sensor - Circuit short to ground or open	NOTE: Circuit reference - WSP RR - Wiring harness or wheel speed sensor connector fault - Wheel speed sensor circuit - Short circuit to ground, open circuit, high resistance - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - Further diagnostics may be accessed using the Jaguar Land Rover Approved Diagnostic Equipment to check datalogger signal - Right Rear Wheel Speed Sensor Input - (0x2B08)
C003A-1C	Right Rear Wheel Speed Sensor - Circuit voltage out of range	NOTE: Circuit reference - WSP RR - Wheel speed sensor circuit - Voltage out of range - Wheel speed sensor circuit - Short circuit to power, short circuit to another sensor, open circuit, high resistance	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to power, short circuit to another sensor, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest
C003A-29	Right Rear Wheel Speed Sensor - Signal invalid	NOTE: Circuit reference - WSS RR - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C003A-2F	Right Rear Wheel Speed Sensor - Signal erratic	NOTE: Circuit reference - WSS RR - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C003A-31	Right Rear Wheel Speed Sensor - No signal	NOTE: Circuit reference - WSS RR - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C003A-4A	Right Rear Wheel Speed Sensor - Incorrect component installed	NOTE: Circuit reference - WSP RR / WSS RR Incorrect wheel speed sensor installed	Check and install the correct wheel speed sensor. Clear the DTCs and retest. To extinguish warning lamps cycle the ignition state to off, then return the ignition state to on and wait up to 30 seconds

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C003A-62	Right Rear Wheel Speed Sensor - Signal compare failure	NOTE: Circuit reference - WSP RR / WSS RR - Wheel speed sensor signal compare failure - Wheel speed sensor circuit - Fault - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C003A-66	Right Rear Wheel Speed Sensor - Signal has too many transitions / events	NOTE: Circuit reference - WSP RR / WSS RR - Unexpected wheel speed sensor signal detected - Wheel speed in excess of maximum expected value - Wheel speed sensor circuit - Fault - Incorrect wheel assembly/tire size - Wheel speed sensor fault	- Check the rear right wheel speed sensor lead and connector for damage - Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C0049-4B	Brake Fluid - Over temperature	NOTE: This DTC will only log on vehicles fitted with carbon ceramic disc brakes The disc wear prediction algorithm has noted prolonged severe brake use that may have resulted in excessive brake fluid temperatures	- Allow sufficient time for the brakes to cool. Refer to the relevant section of the workshop manual and drain/replace the brake fluid as required - Using the manufacturer approved diagnostic tool, check if DTC C1061-68 has been logged by the ABS control module. If this DTC is present, refer to the relevant DTC index for corrective actions - Clear the DTCs and retest
C0061-24	Lateral Acceleration Sensor - Signal stuck high	NOTE: The sensor is located within/integral to the restraints control module Lateral acceleration sensor fault (within restraints control module)	Clear the DTCs and retest. If the fault persists, check the restraints control module for related DTCs and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0061-64	Lateral Acceleration Sensor - Signal plausibility failure	NOTE: The sensor is located within/integral to the restraints control module Lateral acceleration sensor fault (within restraints control module)	Clear the DTCs and retest. If the fault persists, check the restraints control module for related DTCs and refer to the relevant DTC index
C0062-64	Longitudinal Acceleration Sensor - Signal plausibility failure	NOTE: The sensor is located within/integral to the restraints control module Longitudinal accelerometer sensor fault (within restraints control module)	Clear the DTCs and retest. If the fault persists, check the restraints control module for related DTCs and refer to the relevant DTC index
C0063-64	Yaw Rate Sensor - Signal plausibility failure	NOTE: The sensor is located within/integral to the restraints control module Yaw rate sensor fault (within restraints control module)	NOTE: This DTC can be set if the vehicle is being tested on chassis dyno rollers Clear the DTCs and retest. If the fault persists, check the restraints control module for related DTCs and refer to the relevant DTC index
C006B-00	Stability System Active Too Long - No sub type information	- Damaged wheel speed sensor magnetic encoder - Faulty wheel bearing/hub assembly - CAN fault	NOTE: If this DTC is logged for the first time, the only action is to clear the DTC. Should the DTC reoccur, follow the actions as described below - Check integrity of the wheel bearing. Replace as required - Check the integrity of the wheel hub assembly. Replace as required - Use an oscilloscope to examine the signals from all wheel speed sensors for abnormal high frequencies (modulation and peak signals). Renew sensors as necessary - Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network between the yaw rate sensor and the ABS control module; between the steering angle sensor and the ABS control module; and between the lateral acceleration sensor and the ABS control module. Repair circuit as required. Clear the DTCs and retest - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph for at least 30 seconds to ensure the lamps extinguish. When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0072-4B	Brake Temperature Too High - Over temperature	NOTE: This DTC will only log on vehicles fitted with carbon ceramic disc brakes The ABS control module has noted an active braking intervention event has occurred at the same time as at least one brake disc was over temperature	- Allow sufficient time for the brakes to cool. Using the Jaguar Land Rover Approved Diagnostic Equipment, check if DTC C0049-4B has been logged by the ABS control module. If this DTC is present, refer to the relevant DTC index for corrective actions - Using the Jaguar Land Rover Approved Diagnostic Equipment, check if DTC C1061-68 has been logged by the ABS control module. If this DTC is present, refer to the relevant DTC index for corrective actions - Clear the DTCs and retest
C101F-49	Generic Valve Failure - Internal electronic failure	ABS control system hydraulic control unit - Internal fault	- Clear the DTCs and retest. If the fault persists, check and install a new ABS control module, as required - Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)
C105C-68	ABS Dynamometer Test Mode Active - Event information	- Brake pedal, accelerator pedal and the dynamic stability control switch have been simultaneously pressed for at least 5 seconds and the ABS dynamometer test mode has been activated - Dynamic stability control switch stuck	Check the operation of the dynamic stability control switch and rectify as necessary. Clear the DTCs and retest
C1061-06	Brake Disc Wear - Algorithm Based Failures	NOTE: This DTC will only log on vehicles fitted with carbon ceramic disc brakes Disc wear prediction algorithm software failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance - Clear the DTCs and retest. If the fault persists, check and install a new ABS control module, as required - Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1061-68	Brake Disc Wear - Event information	NOTE: This DTC will only log on vehicles fitted with carbon ceramic disc brakes The disc wear prediction algorithm has calculated that one or more of the carbon ceramic brake discs may be worn and in need of replacement	NOTE: If carbon ceramic brake discs are fitted and if one axle set of brake discs is to be replaced, it will be necessary to read the disc wear data from the ABS control module so that the existing wear levels of the unchanged discs may be input into the ABS control module. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance - Check and install new carbon ceramic brake discs as required. Note that carbon ceramic brake discs must be installed in axle sets. Refer to the appropriate removal and installation of the workshop manual. Clear the DTCs and retest - Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the brake discs it is necessary to update the disc wear protection algorithm with the current disc wear levels. If both axle sets of discs are replaced, using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Reset' diagnostic routine (0x0304C). If only one axle set is replaced, using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)
C1109-24	Vehicle Dynamics Control Switch - Signal stuck high	- Dynamic stability control switch stuck - If the switch is pressed for longer than 1 minute, the switch is deemed to be malfunctioning - Switch fault	- Check the operation of the dynamic stability control switch and rectify as necessary. Clear the DTCs and retest - If the fault persists, check and install a new dynamic stability control switch as required
C1A90-12	Wheel Speed Sensor Supply - Circuit short to battery	NOTE: Circuit reference - WSP FL / WSP FR / WSP RL / WSP RR Wheel speed sensor supply circuit(s) - Short circuit to power	NOTE: This DTC is set if at least one wheel speed sensor supply circuit is short circuit to power Refer to the electrical circuit diagrams and check the wheel speed sensor power supply line circuits between the sensors and the ABS control module for short circuit to power. Repair circuit as required. Clear the DTCs and retest
C1A95-01	Wheel Speed Sensor Supply - General Electrical Failure	NOTE: Circuit reference - WSS FL / WSS FR / WSS RL / WSS RR - Wheel speed sensor supply circuit(s) - Short circuit to ground, short circuit to power, short circuit to another sensor, open circuit, high resistance - Battery/charging system fault - Wheel speed sensor(s) fault	- Refer to the electrical circuit diagrams and check the wheel speed sensor ground and power supply line circuits between the sensors and the ABS control module for short circuit to ground, short circuit to power, short circuit to another sensor, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - Refer to relevant section of workshop manual and battery care manual. Check battery state of charge and charging system performance - If the fault persists, check and install new wheel speed sensor(s), as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A95-62	Wheel Speed Sensor - Signal compare failure	NOTE: Circuit reference - WSS FL / WSS FR / WSS RL / WSS RR - Wheel Speed Sensor signal compare failure - Wheel speed sensor fault (any) - Damaged wheel speed sensor magnetic encoder - Faulty wheel bearing/hub assembly - Wiring harness fault - Incorrect wheel assembly/tire size	- Refer to the electrical circuit diagrams and check the each wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - Check integrity of the wheel bearing. Replace as required - Check the integrity of the wheel hub assembly. Replace as required - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle. Check the wheel and tire for condition and balance and rectify/replace as required. Check and tighten the wheel nuts - Use an oscilloscope to examine the signals from all wheel speed sensors for abnormal high frequencies (modulation and peak signals). Renew sensors as necessary. Clear the DTCs and retest - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C1A95-64	Wheel Speed Sensor - Signal plausibility failure	NOTE: Circuit reference - WSS FL / WSS FR / WSS RL / WSS RR - Wheel Speed Sensor signal plausibility failure - Wheel speed sensor fault (any) - Damaged wheel speed sensor magnetic encoder - Faulty wheel bearing/hub assembly - Wiring harness fault - Incorrect wheel assembly/tire size	NOTE: This DTC could be set by a flat tire which would give a continuous different rate of speed when compared with the other 3 wheels. If it is the case, the DTC should be stored as a historic DTC - Refer to the electrical circuit diagrams and check each wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest - If the wheel speed sensor lead and/or connector is damaged or if the fault persists, check and install a new wheel speed sensor as required - Check integrity of the wheel bearing. Replace as required - Check the integrity of the wheel hub assembly. Replace as required - Ensure that the fitted wheels and tires comply to the manufacturers specification for the vehicle. Check the wheel and tire for condition and balance and rectify/replace as required. Check and tighten the wheel nuts - Use an oscilloscope to examine the signals from all wheel speed sensors for abnormal high frequencies (modulation and peak signals). Renew sensors as necessary - After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish - When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A96-64	Brake Light Switch - Signal plausibility failure	■ Signal plausibility failure ■ Stop lamp switch fault ■ Brake pedal switch out of adjustment or faulty ■ Brake pedal switch circuit fault ■ Wiring harness/connector fault	NOTE: This DTC is set when the signal from the brake pedal position switch is contradicted by the signal from the master cylinder pressure sensor ■ Check the stop lamp function. Ensure the stop lamp switch is correctly installed and configured. Check the adjustment of the brake pedal position switch and rectify as required ■ Refer to the electrical circuit diagrams and check the stop lamp switch circuit, for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ■ Refer to the electrical circuit diagrams and check the brake pedal switch circuit, for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ■ Check the engine control module for related DTCs. Refer to the relevant DTC index ■ After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish ■ When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C1A99-14	Pressure Sensor - Circuit short to ground or open	■ ABS control system hydraulic control unit pressure sensor circuit - Short circuit to ground, open circuit, high resistance ■ ABS control module - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance ■ Clear the DTCs and retest. If the fault persists, check and install a new ABS control module as required ■ Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A99-28	Pressure Sensor - Signal bias level out of range / zero adjustment failure	■ ABS control system hydraulic control unit pressure sensor - Signals invalid ■ Pedal box adjustment incorrect or faulty ■ ABS control module - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance ■ Check the operation and adjustment of the pedal box and rectify as required. Clear the DTCs and retest ■ If the fault persists, check and install a new ABS control module as required ■ Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)
C1B00-29	Steering Angle Sensor - Signal invalid	■ Steering angle sensor signal invalid ■ Steering angle sensor not installed in the correct position ■ Steering angle sensor - Failure	■ Check the steering angle sensor installation and orientation. Remove and reinstall if required. Clear the DTCs and retest ■ If the fault persists, check and install a new steering angle sensor as required
C1B00-64	Steering Angle Sensor - Signal plausibility failure	■ Steering angle sensor signal plausibility failure ■ Steering angle sensor installation incorrect ■ Steering angle sensor - Failure ■ Yaw rate sensor installation incorrect	■ This DTC is set when there is conflicting information coming from the steering angle sensor and the yaw rate sensor ■ Check both the steering angle sensor and the yaw rate sensor to ensure they are correctly installed on the vehicle. Remove and reinstall if required. The steering angle sensor is calibrated at every new ignition cycle, every time the vehicle is driven away from standstill. Clear the DTCs and retest ■ If the fault persists, check and install a new steering angle sensor as required
C1B00-76	Steering Angle Sensor - Wrong mounting position	■ Steering angle sensor not installed in the correct position ■ Steering angle sensor - Failure	■ Check the steering angle sensor installation and orientation. Remove and reinstall if required. Clear the DTCs and retest ■ If the fault persists, check and install a new steering angle sensor as required
C1B00-92	Steering Angle Sensor - Performance or incorrect operation	■ Steering angle sensor not installed in the correct position ■ Steering angle sensor - Failure	■ Check the steering angle sensor installation and orientation. Remove and reinstall if required. Clear the DTCs and retest ■ If the fault persists, check and install a new steering angle sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1B02-14	Return Pump - Circuit short to ground or open	NOTE: Circuit reference - PUMP GND ■ Wiring harness fault ■ ABS control module - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance ■ Refer to the electrical circuit diagrams and check the return pump supply fuse, supply circuit and ground circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ■ If the fault persists, check and install a new ABS control module as required ■ Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D) ■ After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish ■ When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C1B02-16	Return Pump - Circuit voltage below threshold	NOTE: Circuit reference - PUMP GND ■ Wiring harness fault ■ Defective fuse	■ Refer to the electrical circuit diagrams and check the return pump supply fuse, supply circuit and ground circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ■ After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish ■ When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1B02-49	Return Pump - Internal electronic failure	NOTE: Circuit reference - PUMP GND ■ ABS hydraulic control unit - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance ■ Clear the DTCs and retest. If the fault persists, check and install a new ABS control module as required ■ Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D) ■ After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish ■ When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C2009-95	Front Axle Wheel Speed Sensors Swapped - Incorrect assembly	■ Vehicle unevenly loaded ■ Incorrect tire(s) installed ■ Wiring harness fault	■ Check the vehicle is not unevenly loaded (heavy on one side). Check that approved wheels and tires are installed. Clear the DTCs and retest ■ If the fault persists, refer to electrical circuit diagrams and check if the front wheel speed sensor connectors are transposed or incorrectly attached at the wiring harness connector. Rectify as necessary. Clear the DTCs and retest ■ After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish ■ When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on
C200A-95	Rear Axle Wheel Speed Sensors Swapped - Incorrect assembly	■ Vehicle unevenly loaded ■ Incorrect tire(s) installed ■ Wiring harness fault	■ Check the vehicle is not unevenly loaded (heavy on one side). Check that approved wheels and tires are installed. Clear the DTCs and retest ■ If the fault persists, refer to electrical circuit diagrams and check if the front wheel speed sensor connectors are transposed or incorrectly attached at the wiring harness connector. Rectify as necessary. Clear the DTCs and retest ■ After rectification, either clear the DTCs and cycle the ignition or cycle the ignition and drive the vehicle above 15kph (the vehicle needs to be driven above 9mph/15kph for more than 10 seconds) to ensure the warning lamps extinguish ■ When cycling the ignition, allow sufficient time for the 'P' symbol illumination to extinguish on the transmission control switch before returning the ignition state to on

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	NOTE: Circuit reference - HS CAN L / HS CAN H - Unexpected data has been received via the high speed CAN network - High speed CAN fault	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network - Using the Jaguar Land Rover Approved Diagnostic Equipment, identify the module(s) responsible for the erroneous CAN signals. Check these modules for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive / sequence counter incorrect / not updated	- Invalid data received from another control module via the high speed CAN bus - High speed CAN fault	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network - Using the Jaguar Land Rover Approved Diagnostic Equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-83	High Speed CAN Communication Bus - Value of signal protection calculation incorrect	- Unexpected data has been received via the high speed CAN network - High speed CAN fault	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network - Using the Jaguar Land Rover Approved Diagnostic Equipment, identify the module(s) responsible for the erroneous CAN signals. Check these modules for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing message	- Unexpected data has been received via the high speed CAN network - High speed CAN fault	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network - Using the Jaguar Land Rover Approved Diagnostic Equipment, identify the module(s) responsible for the erroneous CAN signals. Check these modules for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN fault	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Using the Jaguar Land Rover Approved Diagnostic Equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network
U0074-82	Control Module Communication Bus "B" Off - Alive/sequence counter incorrect/not updated	- Restraints control module fault - Local CAN link between ABS control module and restraints control module - CAN fault	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Clear the DTCs and retest. If fault persists, check the restraints control module for related DTCs and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0074-88	Control Module Communication Bus "B" Off - Bus off	Local CAN link between ABS control module and restraints control module - Bus off	- Refer to the electrical circuit diagrams and check the power and ground connections to the module - Refer to the electrical circuit diagrams and check the private CAN bus circuits between the restraints control module and the ABS control module. Repair circuit as required. Clear the DTCs and retest
U0300-00	Internal Control Module Software Incompatibility - No sub type information	The ABS control module is not compatible with the vehicle	- Check the ABS control module part number, install the correct part as required. Clear the DTCs and retest - To confirm the repair and extinguish the warning lamps, clear the DTCs, cycle the ignition state to off, then return the ignition state to on and wait up to 10 seconds
U201A-55	Control Module Main Calibration Data - Not configured	The ABS control module is not correctly configured	Configure the ABS control module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear the DTCs and retest
U2300-54	Central Configuration - Missing calibration	Configuration missing	Check/amend the car configuration file using the Jaguar Land Rover Approved Diagnostic Equipment
U3000-44	Control Module - Data memory failure	ABS control module memory failure	Clear the DTCs and retest. If the fault persists, contact dealer technical support
U3000-46	Control Module - Calibration / parameter memory failure	- The ABS control module is not correctly programmed - ABS control module - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance - Configure the ABS control module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear the DTCs and retest - If the fault persists, check and install a new ABS control module as required - Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3000-48	Control Module - Supervision software failure	- The ABS control module is not correctly programmed - ABS control module - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance - Configure the ABS control module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear the DTCs and retest - If the fault persists, check and install a new ABS control module as required - Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)
U3000-49	Control Module - Internal electronic failure	ABS control module - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance - Clear the DTCs and retest. If the fault persists, check and install a new ABS control module, as required - Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)
U3000-4B	Control Module - Over temperature	- Module over temperature has been detected - ABS valve overheat protection has been activated	NOTE: This DTC may be set if the Jaguar Land Rover Approved Diagnostic Equipment has been operating the valves for a prolonged period Allow sufficient time for the unit to cool, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3000-51	Control Module - Not programmed	- The ABS control module is not correctly programmed - ABS control module - Failure	NOTE: If carbon ceramic brake discs are fitted and the ABS control module is to be replaced, it will be necessary to read the disc wear data from the original ABS control module before removing it from the vehicle. Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Left/Right Front/Rear Brake Disc Wear Level - (0x2073). See Section 206-09 for further diagnostic guidance - Configure the ABS control module using the Jaguar Land Rover Approved Diagnostic Equipment. Clear the DTCs and retest - If the fault persists, check and install a new ABS control module as required - Note: for vehicles fitted with carbon ceramic brake discs, following the replacement of the ABS control module it is necessary to update the disc wear protection algorithm with the current disc wear levels. Using the Jaguar Land Rover Approved Diagnostic Equipment, run the 'Brake Disc Wear Protection Level' diagnostic routine (0x0304D)
U3000-55	Control Module - Not configured	- Car configuration file incorrect - Central junction box not configured - Central junction box - Failure	- Using the Jaguar Land Rover Approved Diagnostic Equipment check and update the car configuration file as required - Reconfigure the central junction box using the manufacturer approved diagnostic system, clear the DTCs and retest - Reconfigure the ABS control module using the Jaguar Land Rover Approved Diagnostic Equipment, clear the DTCs and retest - If the fault persists, check and install a new central junction box as required
U3003-17	Battery Voltage - Circuit voltage above threshold	Circuit voltage - Above threshold	Refer to relevant section of the workshop manual for battery care requirements. Check the battery state of charge and starting/charging system performance
U3006-16	Control Module Input Power "A" - Circuit voltage below threshold	- ABS control module circuit voltage below threshold - Battery voltage low - Battery ground cable - High resistance - Battery connections loose/corroded - Battery current drain	- Refer to the electrical circuit diagrams and check the module supply fuse, check power and ground circuits for short circuit to ground, open circuit, high resistance. Repair wiring as required, clear the DTCs and retest - Refer to relevant section of the workshop manual for battery care requirements. Check the battery state of charge and starting/charging system performance
U3006-17	Control Module Input Power "A" - Circuit voltage above threshold	NOTE: Circuit reference - PUMP SUPP / VALVE SUPP - ABS control module circuit voltage above threshold - Battery voltage too high - Battery/charging fault	- Refer to the electrical circuit diagrams and check the module supply fuse, check power and ground circuits for short circuit to power. Repair wiring as required, clear the DTCs and retest - Check the engine control module for charging related DTCs and refer to the relevant DTC index - Refer to relevant section of workshop manual for battery care requirements. Check the battery state of charge and starting/charging system performance